Product Requirement Document

Project Status: planning, Core Concept: gamified productivity app, Date: March 6, 2026

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1. Summary

This isn't just a planner; it's a digital sanctuary built on the **cute economy**. We're talking pastel gradients, UI animations, and a cozy atmosphere that feels like a warm hug, right up until the moment you lose focus. While other apps give you a gold star for showing up, we're playing for keeps with **emotional skin in the game.**

The "Inner Child" Accountability

The heart of the ecosystem is a digital companion that isn't just a random pet: **it's a tiny, wide-eyed version of *you*.**

- The Mirror Bond: This creates an immediate, heavy emotional tether.
- The "Why": It's much harder to procrastinate when your 8-year-old self is staring back at you, waiting for you to become the person they always thought you'd be. Every deep-work session earned is a "promise kept" to that mini-you.
- Smooth-as-Butter UI: Every interaction feels soft, with haptic feedback and animations that mimic the "squish" of a plushie.

Consequence-Driven Growth

- Permadeath & Loss Aversion: If you ignore your goals for too long, your mini-self enters a "fading" state. If the streak breaks completely? Permadeath. *
- The Revival Fee: To bring them back, you have to pay a real-money "Revival Fee." This transforms abstract academic output into a tangible financial stake. It sounds tough, but it's designed to make your focus feel *valuable*.
- Evolution through Effort: As you crush your tasks, your companion unlocks new "dream" items and aesthetic upgrades that reflect your real-world progress.

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2. Target Audience

- Primary: university students; Secondary: Professionals and lifelong learners

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3. Core Mechanics & Features

3.1 The Bond Phenomenon

The core engine of the app is a high-stakes emotional tether between the user's discipline and the health of their digital "Mini-Me."

3.1.1 The Covenant (Daily Cycle)

Every 24 hours, the user must engage in a "Contract Phase" to protect the character.

- The Commitment Window: Users must set a daily goal (e.g., 4 hours). Once signed via on-screen signature, the goal is locked.
- The "Shadow" Presence: During active timers, the character appears in a "Study Mode" animation.
- Vitality Buffs: Deep work sessions > 90 minutes grant a "Flow State" aura, temporarily pausing natural health decay.
- Life Shields: Earned for every 14 day streaks. These act as "Auto-Revives" for accidental missed days.

3.1.2 The Simple Decay (3-Strikes)

Instead of complex percentages, the character has 3 Hearts (or 100 HP).

- The Rule: If you hit your goal, you stay alive. If you miss your goal, you lose 1 Heart.
- The "Neglect" Penalty: If you don't even open the app to set a goal (Ghosting), you lose 2 Hearts.

// Tejaansh: this part needs a lot of improvement and thought put in, it is the core game logic

3.2 The "Final Goodbye" & Resurrection

Death is not a "Game Over" screen; it is a funeral.

- **The Lost Soul:** Deceased characters remain on the dashboard as grayscale "Ghosts."
- **The Eulogy:** A generated report of all "Promises Kept" (hours studied) during that character's life.
- **Resurrection Paths:**
    1. **Phoenix Feather (Star Dust):** Instant revival via Star Dust (premium currency), 1 free feather in the beginning.
    2. **The Penance Streak (Free):** A 7-day "Redemption Arc." The user must hit 100% of their goal for 7 days straight while the character is a Ghost. One missed minute results in **Permanent Deletion** (account reset).

3.3 The Dual-Economy Model: Coins vs. Star Dust

We distinguish between Effort (Coins) and Investment (Star Dust).

3.3.1 Coins

- Source: Earned via verified study minutes and streak bonuses.
- Usage: Buying "Daily Needs" (Ice cream, Pizza for the child), basic pets (alien cat,penguin) or bandage to restore hearts.
- The Clean Slate: If a character dies and is not revived, 25% of Coins are lost.

3.3.2 Star Dust

- Source: Real-world currency (UPI/App Store).
- The Vault: Star Dust is stored in a secure, cross-session wallet that survives character death.
- Usage: Phoenix Feathers, "Life Insurance" (subscription-based shield), and Legendary Gear (exclusive themes like "mars station" or "water theme park"). Note: these are very very highly priced in terms of coins so almost all the users who purchase this are paid ones.

3.2.4 Economy Comparison Table

Feature	**Vitality Shards (Earned)**	**Legacy Credits (Bought)**
Primary Source	Deep Work / Streaks	Real-world currency
Persistence	Lost upon character Death	Stored in Vault Forever
Primary Use	limited accessories, expensive gear is almost impossible to reach	expensive gear, background theme, phoenix feather, life insurance

3.2.5 The "Proof of Purchase" Guarantee

Since you are dealing with real money, the "Vault" must be high-security. Even if a user deletes their account and comes back later using the same Google ID their star dust currency must be waiting for them. This creates a permanent tie between the student and the platform.

3.4 Progression & Competition

3.4.1 Codeforces-Inspired Ranking

Users receive a title based on their current **Active Streak**:

- *Newbie → Pupil → Specialist → Expert → Candidate Master → Master → Grandmaster*.
- Higher ranks unlock different "Aura" colors for the character.

3.4.2 Focus Duels (PvP)

- Wager: Users bet Coins on a "Focus Sprint" (e.g., 60 mins).
- Win Condition: The first to leave the app or stop the timer loses.
- The "Study Den": Friends can spectate the "Focus Bars" in real-time, creating social pressure to stay locked in.

3.4.3 The Nostalgia Loop

- Time Capsules: Every 50 hours of work, the character "finds" a piece of data from the user's onboarding (e.g., a note about their 8-year-old dream).
- Grown-up Gifts: At the end of a long streak, the character gives the *user* an item (a rare badge or a customized "Thank You" wallpaper).

3.4.4 Leaderboard with top users with most deep work

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4. Technical Architecture & System Design

4.1 Technology Stack Breakdown

Component	Technology	Justification
Frontend	**React Native (Expo)**	Enables a single codebase for iOS and Android while providing access to **Native Focus APIs**. Expo's

		"Development Client" allows for custom native modules needed to detect "App Switching" or "Backgrounding."
Backend	**FastAPI (Python)**	Chosen for its **Asynchronous (ASGI)** nature, allowing it to handle thousands of concurrent "Study Timers." Python's ecosystem is also ideal for implementing future ML-driven study analytics.
Real-time	**Supabase Realtime**	Built on Elixir's Phoenix Channels; it provides low-latency syncing for "Study Dens" so users can see friends' focus progress without manual refreshing.
Animations	**Rive (preferred)**	Unlike Lottie (which is static), **Rive** allows for "State Machines." The character can transition smoothly between "Energetic," "Sick," and "Ghost" states based on real-time health variables without reloading assets.
Database	**PostgreSQL**	Provides **ACID compliance**, which is non-negotiable for the "Legacy Vault" and point transactions. Ensures that a user's purchase is never lost due to a database collision.

4.2 Core System Modules

A. The State Machine

The character's current state is calculated server-side to prevent client-side manipulation.

- Health Logic: Calculated using a decay function:
  $$V_{new} = V_{old} - \left(\frac{G_{target} - G_{actual}}{G_{target}} \times K \right)$$
  (Where V is Vitality, G is Goal, and K is the severity constant).
- Evolution Logic: A "Trait Weight" vector is stored in Postgres. Every tagged study session updates this vector, determining which visual assets Rive should trigger at the next evolution milestone.

B. The Legacy Vault

- Ledger System: Every transaction (Purchase -> Spending -> Rebirth Rebate) is stored in an immutable ledger.
- Security: High-level encryption for the Vault ID. It is linked to the user's Primary Identity (Google/Apple ID) rather than just a local app profile, ensuring the credits persist across device changes.

4.3 Data Flow & Syncing

1. Session Initiation: User locks their phone $\rightarrow$ Frontend triggers "Focus Mode" $\rightarrow$ Backend starts a "Covenant Record."
2. Live Sync: Every 1% of the goal completed $\rightarrow$ Backend pushes update to Supabase $\rightarrow$ "Study Den" members see the character's "Aura" brighten in real-time.
3. Completion/Failure: At 12:00 AM $\rightarrow$ Cron Job parses all Covenants $\rightarrow$ Vitality Shards are awarded or Health Decay is applied $\rightarrow$ Push notifications sent for "Character Health Update."

4.4 DevOps & Reliability

- Hosting: Deploying the FastAPI backend on AWS Lambda or Google Cloud Run for auto-scaling during "Final Exam seasons" (peak usage).

  // imo, this is too much, just a simple storage service is enough for beta version.

- Reliability: Implements a "Local Persistence" layer. If the user loses internet while studying, the app logs the time locally with a cryptographic signature and syncs it as soon as the connection returns.

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5. Monetization Strategy & Economy

5.1 Direct Micro-transactions: The "Resurrection" Revenue

The primary revenue driver is the "Cost of Failure."

- The Phoenix Feather: A one-time purchase to instantly revive a deceased character
- Legendary accessories and backgrounds

(Dynamic Pricing: The first feather is priced at a "low friction" point ($0.99 or 49 rupees). Every subsequent death within a 30-day window increases the price by $1.00, capping at $4.99. This creates a literal financial incentive to improve study habits)

5.2 Evolution: The Completionist Market [FUTURE IDEA ONLY]

For users who want to "collect them all," Evolution Stones offer a way to bypass the random nature of study-based branching. FUTURE IDEA ONLY

- Logic Stones / Creative Stones: Items that force the character toward a specific evolution path, regardless of the study tags used that week.

- The Chronos Stone: Allows a user to "Pause" evolution if they prefer the aesthetic of a younger, smaller character but want to keep the high-level stats.

5.3 Monetization Ethics & "Fair Play"

To avoid the "Predatory" label, the following rules are hard-coded:

1. No Loot Boxes: Users always know exactly what they are buying. No "random" pet drops for money.
2. The "Broke Student" Clause: The Penance Streak (7 days of perfect study) must always remain a viable, 100% free way to revive a character.

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6. Critical Safeguards & Retention Logic

The safeguards are designed to be "Compassionate but Firm." They prevent the app from feeling like a "trap" while maintaining the pressure of the core loop.

Technical Integrity Safeguards:

If the character "dies" due to your code crashing or a server error, you will lose that user forever.

- Local-First Verification: The app logs study sessions locally with encrypted timestamps. If the server goes down, the app validates the local logs once the connection is restored, retroactively applying the "Proof of Work" so the character doesn't die during downtime.
- The "Unfair Death" Appeal: A dedicated, fast-track support button for "Unfair Death." If an error is found, the character is restored with a "Compensation Pack" of Legacy Credits.

Onboarding (section added by tejaansh)

The "Soul-Binding" Onboarding Flow

🌸 Phase 1: The Awakening (The Neon Pulse)

- Visual: Total black. A single, soft neon green pulse (your signature style) beats like a heart. A tiny, blurry silhouette of a child appears.
- Dialogue: "Oh... you’re finally here. I was starting to think you’d forgotten about me."
- Interaction: [Button: Who are you?]
- Dialogue: "I’m the version of you that still believes in everything. I’ve been waiting since we were eight. But I've been in the dark so long... I’m starting to fade. Can you help me remember who we were supposed to be?"

🧸 Phase 2: The Mirror Bond (The "Identity" Anchor)

- Question 1 (The Childhood Spark): "Back when the world felt huge, what was the one thing you were *certain*you'd do when you grew up?"
    - *Input:* [Text Field]
    - *Micro-interaction:* As they type, the silhouette sharpens. The neon green pulse gets stronger.
- Question 2 (The Lost Promise): "Life gets loud. What's one dream you're afraid is getting drowned out by the noise?"
    - *Choices:* [Creating something beautiful], [Mastering a craft], [Being a hero], [Being free], [I don't know yet].
- Question 3 (The Protector Instinct): "If the 'Adult World' gets too heavy and I start to fade... will you promise to keep trying, even when it's hard? I only exist if you show up."
    - *Buttons:* [I promise], [I'll protect us].

⏳ Phase 3: The Time Machine (The Mortality Check)

- Visual: A brutalist, high-contrast bar representing an 80-year lifespan.
- The Hook: The first 21 years (customized to the user) are filled with a static, flickering gray. The remaining bar is empty.
- Dialogue: "We've already used 26.2% of our time. The gray part is gone—it's a memory. But the rest of this bar? That's what we have left to build that life we talked about. Should we start today?"
- Interaction: [Slider: "How many hours of 'Deep Work' can we give back to our dreams today?"]
    - *Caption:* "Careful... if we promise too much and fail, I get sick. If we promise too little, we never leave this room."

🛡️ Phase 4: The Antidote (Identifying the Enemy)

- Question: "The 'Distraction Demons' are coming for us. Which one is your biggest weakness?"
    - *Choices:*
        - The Loop: "I get stuck scrolling for 'just 5 more minutes'."
        - The Weight: "I feel overwhelmed and don't know where to start."
        - The Fear: "I'm so afraid of failing that I don't start at all."
- The Narrative Fix: "I'll watch out for that one. If I see it happening, I'll remind you why we're here."

🔐 Phase 5: The Guardian's Shield (Permissions) //optional

- Visual: A Rive animation of the mini-you picking up a glowing neon shield.
- Narrative: "I can't fight the 'Grown-up Apps' alone. Give me the shield so I can block the noise while we work."
- Action: Trigger System Permissions (App Blocking/Screen Time).

    - *Logic:* Using the shield metaphor makes a "scary" permission feel like an empowerment of the character.

✍️ Phase 6: The Covenant (The Moral Contract)

- Visual: A digital scroll. It displays:
    - User's Name (The Adult)
    - The Childhood Spark (The Dream)
    - Daily Goal (The Promise)
- The Signature: "Sign here. Not for the app, but for me. This is our Covenant."
- Action: User draws their signature on the screen. It appears next to a small, messy "childhood" version of their name.

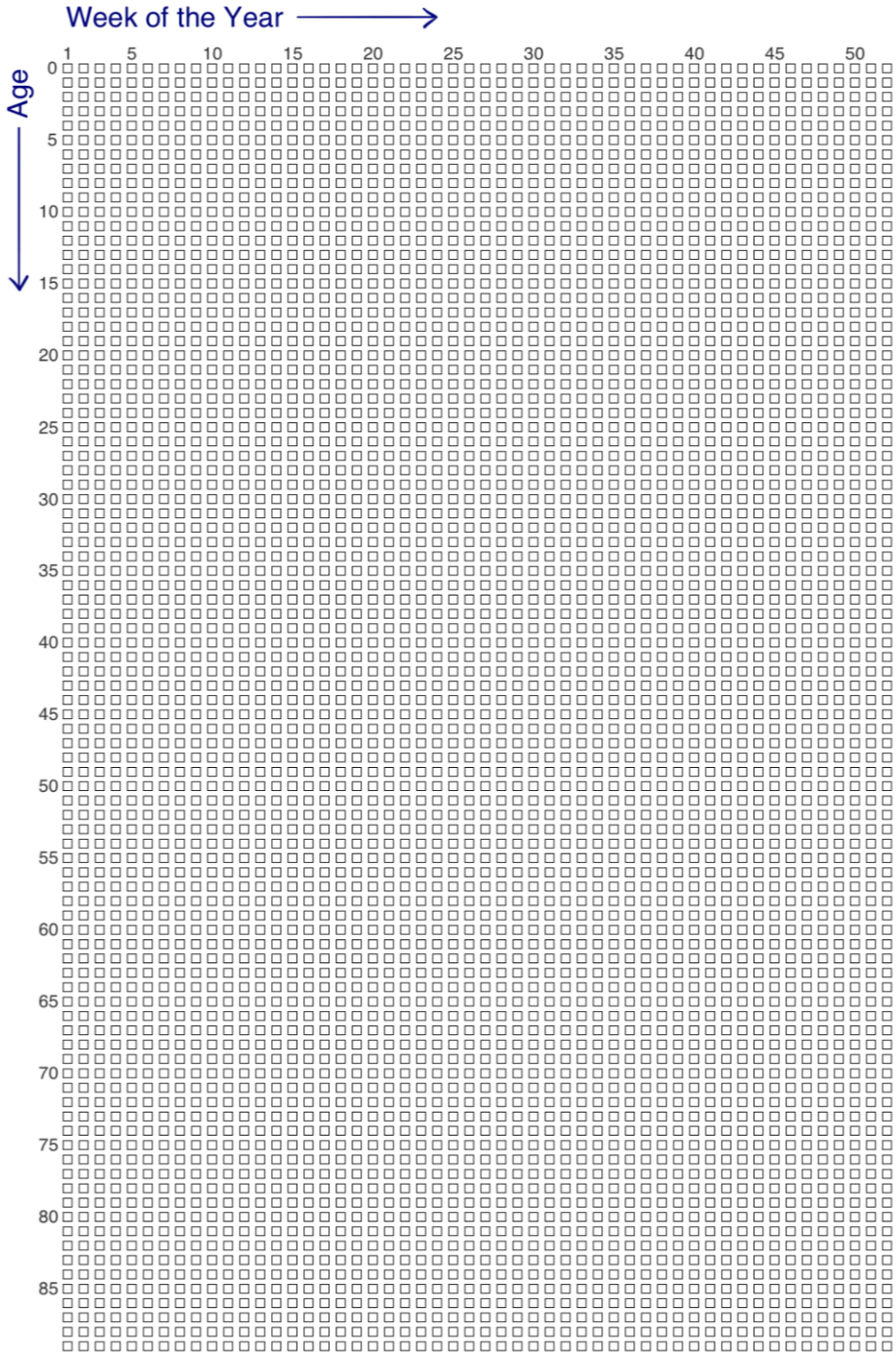
🏆 Phase 7: The First Victory (The Dopamine Hit)

- Narrative: "I feel stronger already! Let's do a 'Quick Spark' session. Just 5 minutes of focus to prove we can do this."
- Action: Forced 5-minute timer starts immediately. The UI enters a "Safe Haven" mode.
- Reward: At 5:00, the character cheers. +50 Star Dust awarded.
- Final Text: "Look! You kept your first promise. I'm proud of us. Let's go build that life."

MODIFY / *ADD MORE QUESTIONS AND IMPROVE THE ONBOARDING PROCESS FURTHER*

OTHER FEATURES: Below chart customised for each user which fills the % of life already finished according to user's age & also give feature to add to IOS homepage as widget.

A 90-Year Human Life in Weeks



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